

The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Comparative Study, with a Mechanism Analysis | Author: Zihao Huang | October 2026

Abstract

This study builds a simplified single-aisle cabin boarding model to compare the total boarding time of three strategies: *random*, *back-to-front*, and *zoning (5 zones)*. Contrary to intuition, the simulation shows that **random boarding is the fastest and back-to-front is the slowest**, and the gap widens as stow time and the number of rows increase. Beyond measuring the times, this study **instrumented the model to expose the underlying mechanism**: it records the average number of passengers stowing luggage *simultaneously* (parallelism) and the total number of *blocking events*. The results show that the fastest strategy is also the one with the highest parallelism and the fewest blocking events, quantitatively explaining *why* it is faster.

1. Background and Research Question

Boarding time strongly affects on-time performance and operating cost. Airlines often adopt "zone boarding" or "back-to-front boarding", but whether these outperform random boarding is debated. This study tests the following hypothesis:

Hypothesis H: Organized strategies (back-to-front / zoning) board faster than random boarding.

2. Methods

2.1 Model

- The cabin has a **single aisle**; passengers enter from the front and walk to their assigned row.
- On reaching their row, a passenger **stows luggage**, occupying the aisle for **STOW** ticks; followers cannot pass.
- Each passenger advances at most one cell per tick; if blocked, they wait (no overtaking).
- Only **one passenger enters** per tick (single-file boarding).
- The simulation ends when all passengers are seated; the total number of ticks is the **boarding time** (smaller = faster).

2.2 Variables

Type	Variable
Independent	Boarding strategy (Random / Back-to-front / Zones-5)
Dependent	Total boarding time (ticks)
Control (baseline)	Random boarding (normalized to 1.00)
Controlled	Rows R; seats per row (6); entry rate (1/tick); repeats (20)

2.3 Mechanism Metrics (key methodological addition)

To explain the results rather than merely report them, the simulator also records, at every tick:

- **Concurrency:** how many passengers are stowing luggage at the same time (a measure of *parallelism*). Reported as the average over the whole boarding process.
- **Blocking events:** how many walking passengers are stuck because the cell ahead is occupied (a measure of *congestion*).

2.4 Data Collection

Strategies with randomness (Random, Zones) were each run **20 times and averaged** to reduce random error; Back-to-front is deterministic. Results are reported relative to random (baseline = 1.00).

3. Results

3.1 Main Result (R = 30 rows, STOW = 3)

Strategy	Average boarding time (ticks)	Relative to random
Random	331.9	1.00 (baseline)
Zones (5)	402.6	1.21
Back-to-front	627	1.89

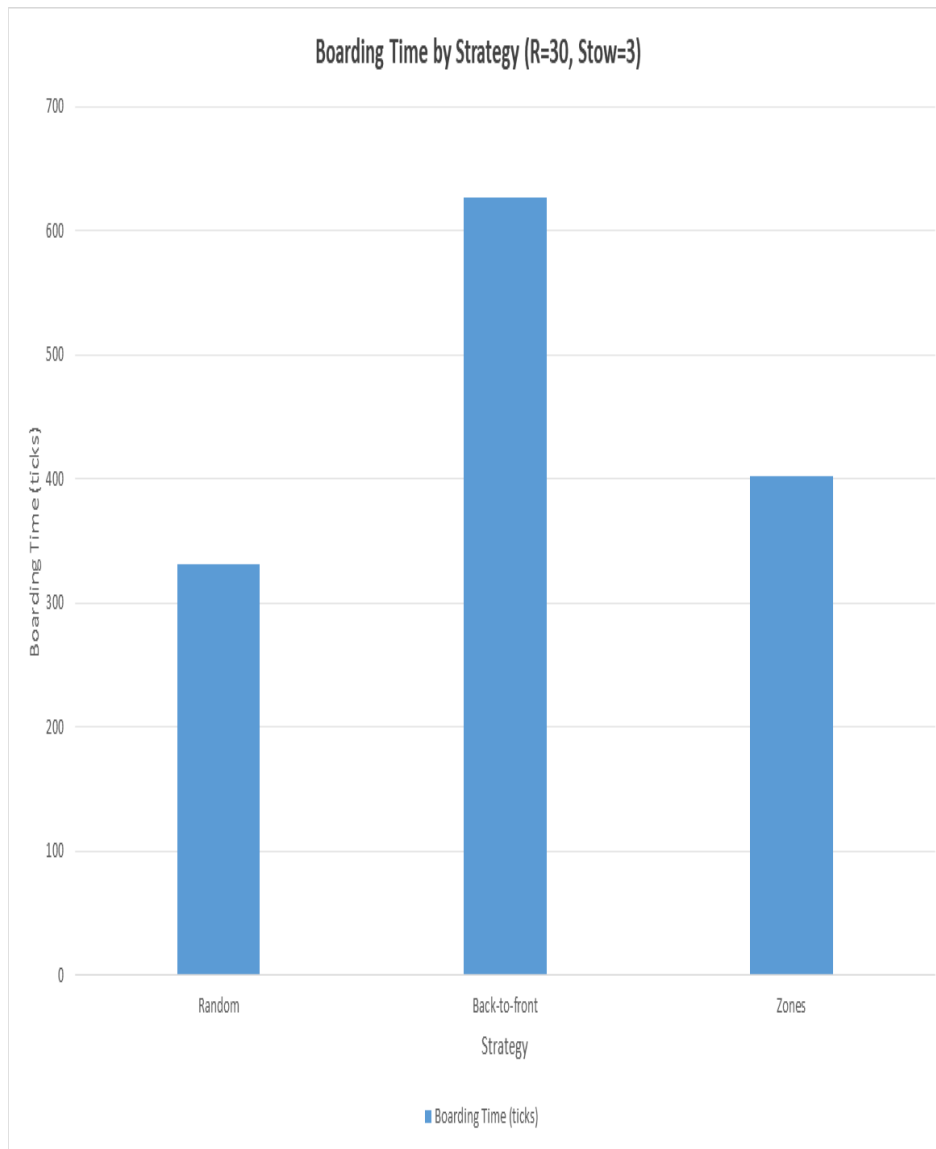


Figure 1. Boarding time by strategy (R=30, STOW=3): random is fastest, back-to-front slowest.

3.2 Sensitivity Analysis

(a) Varying stow time (fixed R = 30). The disadvantage of back-to-front grows sharply with stow time.

STOW	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
1	238.9	325	249.9	1.36	1.05
3	331.9	627	402.6	1.89	1.21
6	471.5	1080	633.8	2.29	1.34

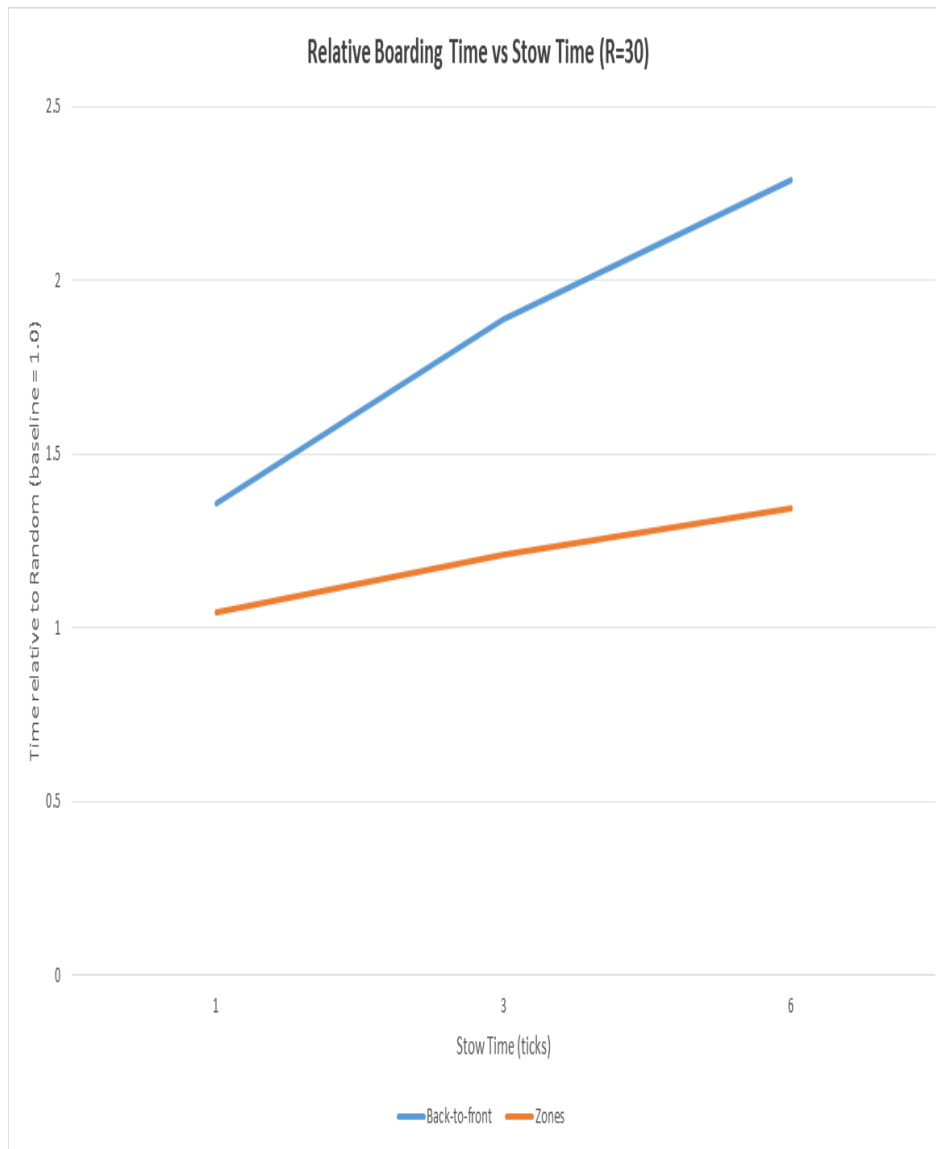


Figure 2. Relative boarding time vs stow time (R=30): back-to-front worsens steadily; zoning stays nearly flat.

(b) Varying number of rows (fixed STOW = 3). Larger aircraft penalize back-to-front more.

Rows R	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
20	234.3	417	288.7	1.78	1.23
30	331.9	627	402.6	1.89	1.21
40	423.6	837	507.3	1.98	1.20

3.3 Mechanism Analysis (why the times differ)

Averaging over repeated runs, the mechanism metrics for the three strategies are:

Strategy	Boarding time (ticks)	Avg concurrent stowers (parallelism)	Total blocking events (congestion)
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Random	329	1.09 (highest)	~950 (fewest)
Zones (5)	401	0.89	~3,000
Back-to-front	627	0.57 (lowest)	6,525 (most)

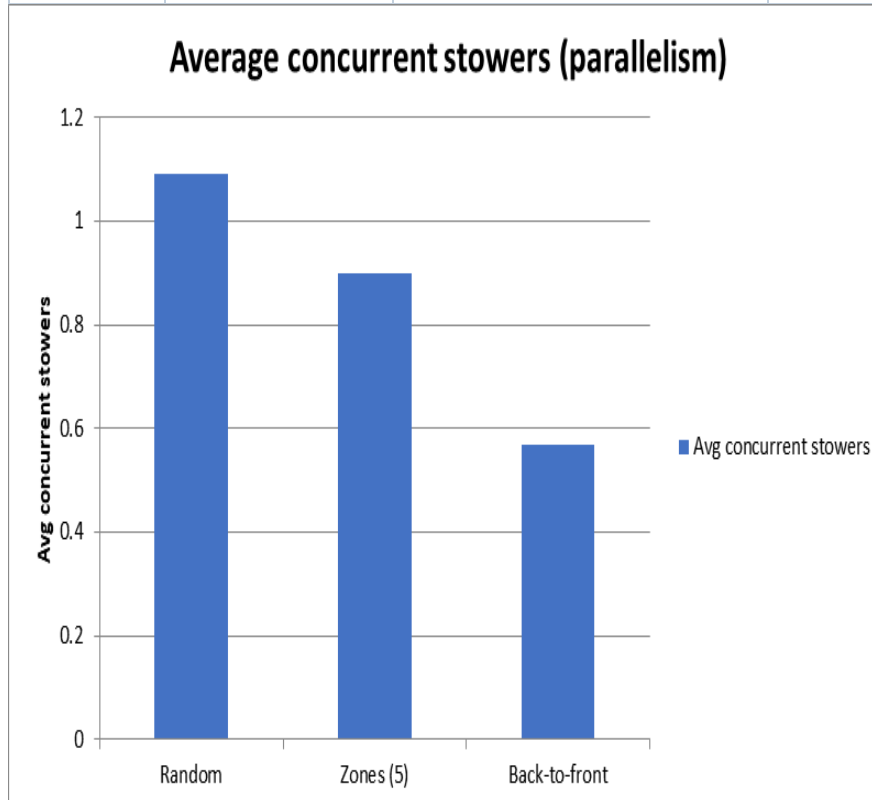


Figure 3. Average concurrency (parallel stowing) by strategy: random is the most parallel.

The metrics are mutually consistent with the times: the fastest strategy (random) has the **highest parallelism** (about 1.09 passengers stowing at once) and the **fewest blocking events** (about 950), whereas the slowest (back-to-front) has the **lowest parallelism** (0.57) and roughly **7× more blocking events** (6,525). This provides direct evidence for the proposed mechanism rather than only a qualitative claim.

4. Discussion

Why is random fastest? Random boarding spreads passengers across rows, so several rows stow luggage **simultaneously** — the measured parallelism is highest (1.09). More parallel stowing means fewer people are blocked waiting.

Why is back-to-front slowest? It boards the six passengers of the same row consecutively; they must stow at the same aisle position, so they proceed **one by one**. The measured parallelism is the lowest (0.57) and blocking events are the highest (6,525).

Consistency with the sensitivity analysis: blocking is driven by stow time; hence the disadvantage of back-to-front grows with STOW (Figure 2). This causal pattern — more stow time → more blocking → larger gap — supports the mechanism.

5. Limitations and Future Work

- The model is simplified: it ignores assigned-seat effects, dual aisles, two doors, and passengers stepping into seats immediately after stowing.
- Concurrency is capped by the model's single-aisle assumption; real cabins may allow slightly more overlap.
- Future work could add "window-middle-aisle" and the "Steffen method", and test whether disabling parallelism removes the strategy differences (a direct causal test).

6. Conclusion

In this simulation, **dispersed boarding (random) outperforms clustered boarding (back-to-front)**. Beyond measuring the times, instrumenting the model shows the cause: random boarding yields the **highest parallelism** in luggage stowing and the **fewest blocking events**, whereas back-to-front yields the lowest parallelism and the most blocking, and its disadvantage grows with stow time and cabin size. The findings are relevant to the design of boarding strategies in airline operations.

References

Steffen, J. H. (2008). Optimal boarding method for airline passengers. *Journal of Air Transport Management*. Van Landeghem, H., & Beuselinck, A. (2002). Reducing passenger boarding time in airplanes. *European Journal of Operational Research*. Bachmat, E., et al. (2006). Analysis of airplane boarding via space-time geometry and random matrix theory.